

Design of a Low Cost, High Precision Laser Wavemeter using a Fizeau Wedge Interferometer

By

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Abstract

This work presents the design, construction, and experimental validation of a low-cost laser wavemeter based on a Fizeau wedge interferometer. Commercial wavemeters capable of sub-nanometer wavelength accuracy typically cost several thousand dollars, limiting their accessibility in undergraduate teaching and research environments [1]. The system developed here addresses this gap by combining a fused silica Fizeau wedge, a Raspberry Pi HQ CMOS camera, and a Raspberry Pi 5 to achieve real-time wavelength measurement at a total component cost of approximately \$972.87 CAD.

A collimated laser beam is directed onto the wedge at a fixed incidence angle of 22° , producing spatial interference fringes at the zero-shear plane. A Python analysis pipeline running on the Raspberry Pi extracts a one-dimensional intensity profile from the captured fringe image, applies a Fast Fourier Transform (FFT) to identify the dominant spatial frequency, and recovers the incident wavelength using the known wedge geometry and refractive index. An optical ray model and Python simulations were used to characterize zero-shear plane position and validate the FFT-based wavelength extraction method prior to experimental testing.

The wavemeter was tested with two laser sources. For a 532 nm green Nd:YAG laser, a wavelength of 532.48 nm was measured, giving a discrepancy of 0.48 nm from the theoretical value. For a red diode laser characterized at 653.07 ± 0.001 nm by spectrometer, the wavemeter returned 654.19 nm, a discrepancy of 1.12 nm. Pre-calibration systematic uncertainties, primarily wedge angle tolerance and pixel pitch variation, are identified as the dominant error sources. The results demonstrate sub-nanometer measurement capability approaching the ± 0.1 nm design target, with clear pathways for further improvement through sub-bin FFT interpolation techniques and LabVIEW integration.

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1 Introduction

1.1 Motivation

High-precision measurement of laser wavelengths is fundamental to optics research and spectroscopy [2]. However, commercial wavemeters capable of nanometer or picometer accuracy typically cost several thousand dollars, placing them out of reach for many undergraduate teaching spaces and researchers [1]. As a result, exposure to wavelength metrology remains limited, creating barriers for students and researchers interested in optical science and engineering.

Improving accessibility to optical instrumentation is therefore a key motivation for this work. By reducing cost and complexity, a wider range of users can engage with precision optical measurements.

1.2 Problem Definition

Despite the availability of high-performance commercial wavemeters, their cost and complexity make them impractical for many instructional and resource-constrained environments. This project addresses the development of a wavelength measurement system that achieves sub-nanometer measurement uncertainty while remaining affordable, and mechanically simple. This work focuses on designing a Fizeau-based system capable of extracting wavelength information from a single interference pattern, while maintaining simplicity and ease of use.

1.3 Project Objectives

The primary objective of this project is to design and implement a low-cost wavemeter capable of accurately measuring tunable diode lasers with wavelengths in the range of 600–900 nm and output powers of 1–100 mW, with a measurement uncertainty of ± 0.1 nm.

The system uses a Fizeau interferometer architecture, identified in prior work as a simple and cost-effective approach for wavelength measurement [3]. The device captures interference fringe

patterns using a CMOS camera and extracts wavelength information through computational analysis. This approach eliminates the need for moving components, improving mechanical stability, reducing maintenance, and enabling a compact design.

Another key objective of this project is maintaining a total system cost under \$1500 CAD. This budget is intended to ensure that the wavemeter remains accessible for education and research groups while still achieving meaningful performance.

1.4 Application Context

The laser wavelengths and power operating range is specifically selected to align with the tunable diode lasers used in Dr. Stotz's laboratory. The target measurement uncertainty of 0.1 nm wavelength is motivated by GaAs quantum-well spectroscopy applications in Dr. Stotz's laboratory, where optical transition linewidths are on the order of 1 nm. Achieving 0.1 nm resolution enables precise and repeatable tuning of diode lasers within a small fraction of the transition linewidth, ensuring reliable excitation of quantum states and supporting accurate measurements.

1.5 Constraints

To guide the design of the wavemeter, a set of constraints was established to balance performance, cost, and usability, with particular consideration for its intended application in Dr. Stotz's research. Table 1 summarizes the requirements that the system must meet.

Table 1. Design constraints for a low-cost laser wavemeter, specifying performance, cost, mechanical, and usability requirements.

Category	Constraint	Description
Cost	Total system cost < \$1500 CAD.	Includes optics, mounts, CMOS camera, electronics, and mechanical components.

Category	Constraint	Description
Wavelength Performance	Measurement uncertainty = 0.1 nm.	Chosen because GaAs quantum-well transitions have linewidths of roughly 1 nm; a 0.1 nm resolution ensures the wavemeter can tune diode lasers well within that linewidth for reliable excitation. Relevant to Dr. Stotz research.
Laser Input Range	1–100 mW diode lasers.	Optical elements and the camera must tolerate this range without damage or use optical attenuation to meet camera pixel saturation specifications.
Camera Constraints	>10Hz Refresh, No Infrared filter (NoIR), and High Pixel Resolution.	A refresh rate above 10 Hz enables real-time wavelength readings during laser tuning. A NoIR sensor extends sensitivity across a wider wavelength range. The pixel resolution will be determined through Python modelling to ensure the camera can accurately resolve the fringes.
Optical Geometry	Static, no moving parts.	Only single-frame fringe capture is allowed; no scanning stages or reference lasers.
Mechanical Stability	Robust against minor misalignment.	Designed for repeated handling in undergraduate labs. Must be tunable.
Portability	Fits on a small optical breadboard.	Device must be easy to transport across labs and classrooms.

Category	Constraint	Description
Thermal Stability	Minimal performance drift with small temperature changes.	Wedge angle and alignment should remain stable in typical lab environments.
User Accessibility	Simple alignment and operation.	Suitable for users with limited optics experience; minimal adjustment required.

1.6 Risks and Safety Considerations

The wavemeter is intended for use with visible and near-infrared diode lasers in the 1–100 mW range, placing the system in the Class 3B laser category [4]. Operation of such lasers requires adherence to Queen’s University laser safety protocols, including the use of appropriate protective eyewear, controlled laboratory access, and clear signage indicating the presence of higher-power laser equipment [5].

Although the optical design of the Fizeau interferometer inherently distributes a portion of the incident beam through reflections within the wedge, the optical power reaching the CMOS sensor must remain within safe operating limits [6]. In this implementation, safe operation is achieved by controlling laser power, maintaining appropriate beam alignment, and using the camera within its specifications to prevent sensor saturation.

To further mitigate the risk of excessive optical intensity in higher-power configurations, optical attenuation methods such as neutral density filters or beam pick-off arrangements may be incorporated. While these were not required for the present experimental setup, they provide a scalable approach for adapting the system to higher-power laser sources.

Mechanical risks associated with alignment are primarily related to unintended beam reflections during setup. These are mitigated by performing alignment procedures at low laser power, using beam blocks, and minimizing reflective surfaces in the optical path. Thermal and electrical risks

are minimal because the Fizeau wedge and mounts operate passively with no heating elements, and the electronics will run at low voltages [7].

2 Background and Theory

2.1 Coherence and Beam Properties

The formation of interference fringes depends on several properties of the incident light, including coherence, collimation, and spectral bandwidth. While collimation determines spatial direction and bandwidth describes wavelength spread, phase coherence fundamentally governs the ability to produce interference patterns [8]. These properties are illustrated in Figure 1, which compares different combinations of coherence, collimation, and spectral content.

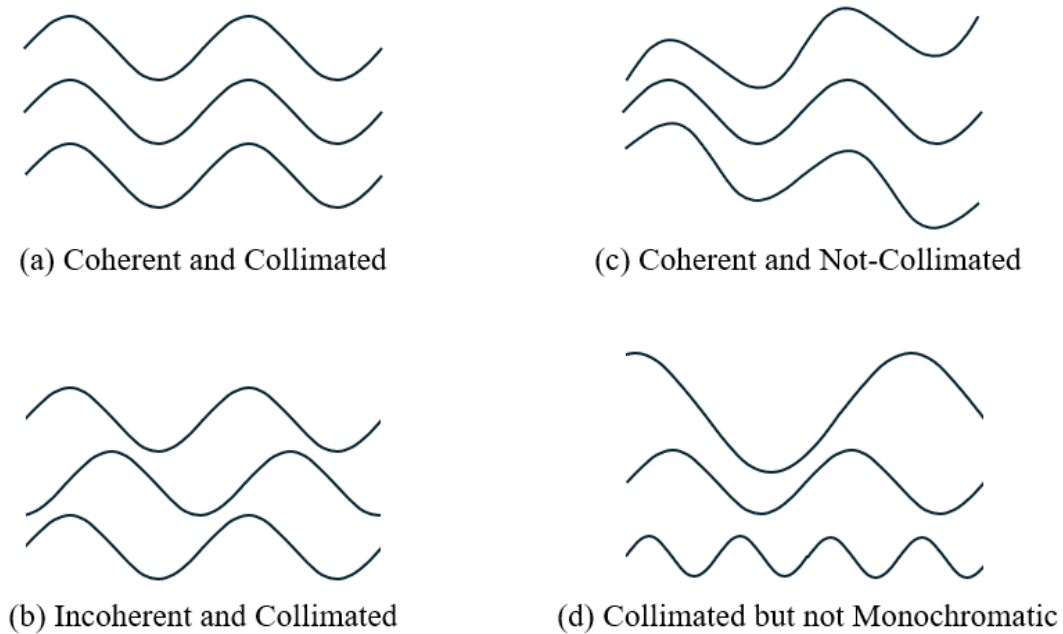


Figure 1. Illustrates different combinations of coherence, collimation and spectral content.

Interference fringes require that the light source maintain a stable phase relationship over the optical path difference (OPD) between interfering beams. This requirement is quantified by the coherence length, defined as the distance over which the phase correlation of the wave is

preserved. Interference is only observed when the OPD is less than or comparable to the coherence length [9].

The coherence length is inversely related to the spectral bandwidth $\Delta\lambda$, and can be written as:

$$L_c = \frac{\lambda^2}{\Delta\lambda} \quad (1)$$

indicating that narrow-linewidth sources produce longer coherence lengths.

For diode lasers, the coherence length is typically much larger than the OPD within a Fizeau wedge, ensuring high-contrast interference fringes. While finite bandwidth can slightly reduce fringe visibility and introduce uncertainty in fringe spacing, this effect is negligible for narrowband laser sources [10].

2.2 Interference Fundamentals

Interference occurs when coherent waves superpose to produce an intensity pattern dependent on their phase difference [11]. When two waves of the same wavelength λ travel different optical path lengths, the resulting phase difference is given by:

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta d \quad (2)$$

where Δd is the OPD. Constructive interference occurs when $\Delta\phi = 2\pi m$, and destructive interference occurs when $\Delta\phi = (2m + 1)\pi$, for integer m [11]. Since the phase difference is inversely proportional to wavelength, interference patterns provide a sensitive method for measuring small changes in wavelength.

2.3 Fizeau Wedge Interferometry

A variety of optical techniques exist for measuring laser wavelength, including diffraction-based methods and interferometric approaches. A comprehensive evaluation of these techniques was conducted in prior work [3], which identified the Fizeau wedge interferometer as the most suitable architecture for a low-cost, and compact wavemeter design.

A Fizeau wedge interferometer consists of a transparent wedge with a small angle between its surfaces. When a collimated beam is incident on the wedge, partial reflections from the front and back surfaces interfere, producing a spatial fringe pattern [12]. The local thickness of the wedge varies linearly with position x , leading to a position-dependent OPD $\Delta d(x)$. The resulting intensity pattern can be expressed as [12]:

$$I(x) = I_0 \left[1 + \cos \left(\frac{2\pi}{\lambda} \Delta d(x) \right) \right] \quad (3)$$

For small wedge angles, the fringe spacing Δx is:

$$\Delta x = \frac{\lambda}{2n\alpha} \quad (4)$$

where n is the refractive index and α is the wedge angle. This relationship forms the basis for wavelength extraction in Fizeau wavemeters.

2.4 Refractive Index and Dispersion Effects

The refractive index of the wedge material plays a critical role in determining the interference pattern, as it directly appears in the relationship between fringe spacing and wavelength (Equation 4). As a result, accurate wavelength extraction requires knowledge of the refractive index of the material.

For optical materials such as fused silica, the refractive index is not constant but varies with wavelength, a phenomenon known as material dispersion [13]. This dependence must be accounted for when converting measured fringe spacing into wavelength, particularly over broad spectral ranges.

The wavelength dependence of the refractive index is described using models such as the Sellmeier equation [13], given by:

$$n(\lambda) = \sqrt{1 + \frac{0.6961663\lambda^2}{\lambda^2 - 0.0684043^2} + \frac{0.4079426\lambda^2}{\lambda^2 - 0.1162414^2} + \frac{0.8974794\lambda^2}{\lambda^2 - 9.816161^2}} \quad (5)$$

Using this relation, the variation of refractive index with wavelength for fused silica at room temperature (293 K) is shown in Figure 2. Over the wavelength range relevant to this work, the refractive index varies gradually, but this variation is sufficient to introduce measurable error if neglected.

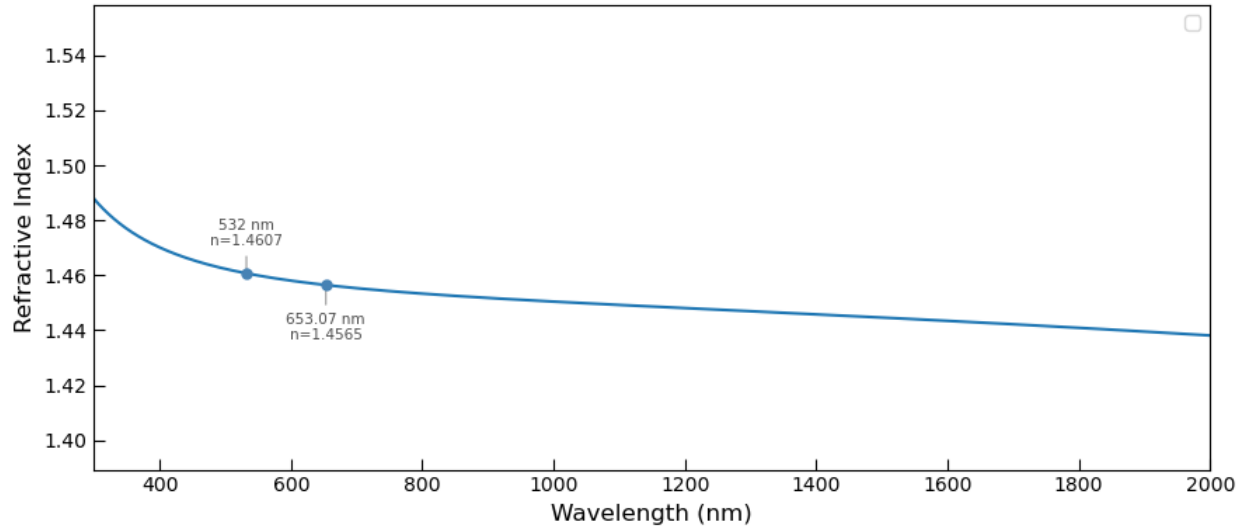


Figure 2. Illustrates the variation of refractive index (n) with wavelength for fused silica at room temperature (293 K). This figure was generated in Python using the Sellmeier coefficients for fused silica [13]. A wavelength of 532 nm and 653.07 nm is shown as examples.

2.5 Computational Analysis of Interference Patterns

Modern wavemeters rely on digital image processing techniques to extract wavelength information from interference patterns [1]. Two primary computational approaches are considered in this work.

The first approach uses an FFT based fringe-spacing method. In this approach, a one-dimensional (1D) intensity profile is extracted from the interference image and transformed into the frequency domain using a Fourier transform. The dominant spatial frequency of the fringes is then identified and related to the wavelength through the wedge geometry, as described by Equation 1 [14].

The second approach is a spatial peak-fitting method, based on the high-precision interferometric technique demonstrated by *Singh et al.* [15], and summarized in Figure 3. Here, a 1D intensity profile is similarly extracted from the interference image, and the positions of intensity maxima and minima are identified. These positions are plotted against fringe number and fitted with a linear model, where the slope corresponds to the fringe spacing and the intercept defines the zero-fringe position. Using the known wedge geometry and refractive index, the wavelength is then determined from the measured fringe spacing, as described by Equation 1, with a final linear calibration applied.

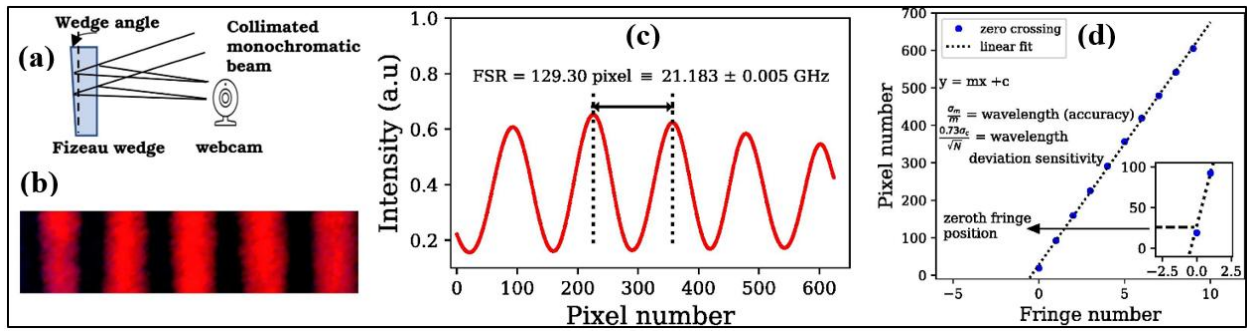


Figure 3. Overview of the spatial peak-fitting method. (a) Optical layout of the Fizeau wedge wavemeter. (b) Example raw fringe image captured by the camera. (c) Extracted 1D intensity profile showing the free spectral range (FSR) between adjacent maxima. (d) Pixel number plotted against fringe number with a linear fit, from which the refined fringe spacing and phase shift are obtained. The slope and intercept of this fit provide a highly sensitive measure of wavelength, enabling sub-nanometer resolution. Reproduced from [15].

3 Methodology

3.1 Optical Setup and Fringe Formation

The wavemeter operates by generating and analyzing interference fringes produced by a solid Fizeau wedge. Figure 4 below illustrates the optical path travelled by the laser beam. A collimated laser beam is directed onto the wedge at a fixed incidence angle. Upon reaching the upper surface of the wedge, a portion of the beam is reflected while the remainder is transmitted into the material. The transmitted beam reflects off the lower surface of the wedge and exits the

material, where it interferes with the beam reflected from the upper surface. Due to the varying thickness of the wedge, the OPD between these two beams changes as a function of position, producing a spatial interference pattern. This pattern consists of alternating bright and dark fringes corresponding to constructive and destructive interference [12].

The resulting interference fringes will be projected onto a CMOS camera positioned at the zero-shear plane, where the two beams intersect. The captured fringe pattern encodes the wavelength of the incident laser through its spatial frequency.

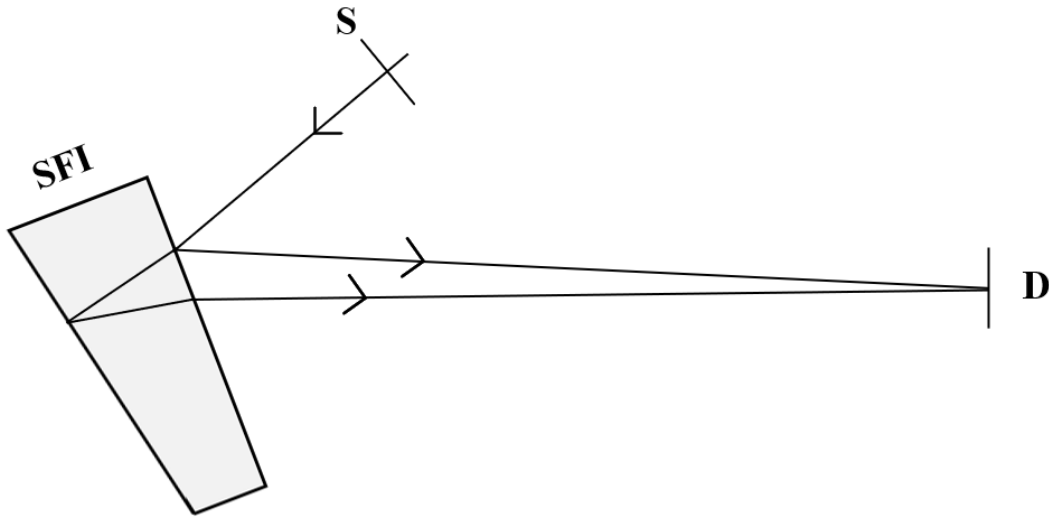


Figure 4. An incident light beam strikes the upper surface of the solid Fizeau interferometer (SFI), where it is partially reflected and partially transmitted into the wedge. The transmitted beam reflects off the lower surface and refracts out of the wedge, overlapping with the beam reflected from the upper surface. The two beams interfere at the detector (D) to produce interference fringes. Adapted from the model presented in [12].

3.2 Image Acquisition

The interference pattern is captured using a CMOS camera with a fixed-focus configuration, ensuring that the imaging plane remains constant and aligned with the plane of the interference fringes. This prevents variations in focus that could obscure the fringe pattern. To optimize fringe visibility, the incident beam will be spatially filtered and expanded to produce a near-

Gaussian intensity profile. Alignment will be performed at low power to ensure that the interference pattern is centered on the detector and spans a sufficient number of pixels for accurate analysis.

3.3 Computational Processing and Wavelength Extraction

Captured images will be processed using a Python analysis pipeline executed on a Raspberry Pi, which serves as the computer for the wavemeter. The Raspberry Pi processes acquired images via a ribbon cable connection to the camera, eliminating the need for an external computer. Development and execution of the program are performed by remotely connecting to the Raspberry Pi via Secure Shell (SSH), allowing the user to run and modify the Python analysis pipeline and retrieve results such as fringe images and plots.

Each captured image will be converted into a 1D intensity profile by extracting a slice through the center of the interference pattern, perpendicular to the fringes. The resulting signal represents the spatial variation in intensity associated with the interference fringes.

To isolate the fringe signal, zero-frequency background contributions are suppressed by removing the low spatial frequencies. The fringe spacing will then be determined using an FFT, where the spatial frequency spectrum of the intensity profile is computed and the dominant frequency component is identified. This dominant spatial frequency will be used to calculate the fringe spacing and subsequently recover the wavelength of the incident laser. The recovered wavelength will then be displayed in real-time on a small display interfaced with the Raspberry Pi.

4 Calculations, Models and Analysis

4.1 Optical Ray Model

The optical ray model for this Fizeau wedge is illustrated in Figure 5, showing the incoming laser beam striking point A, then reflecting and refracting until it reaches the plane of zero shear at point D.

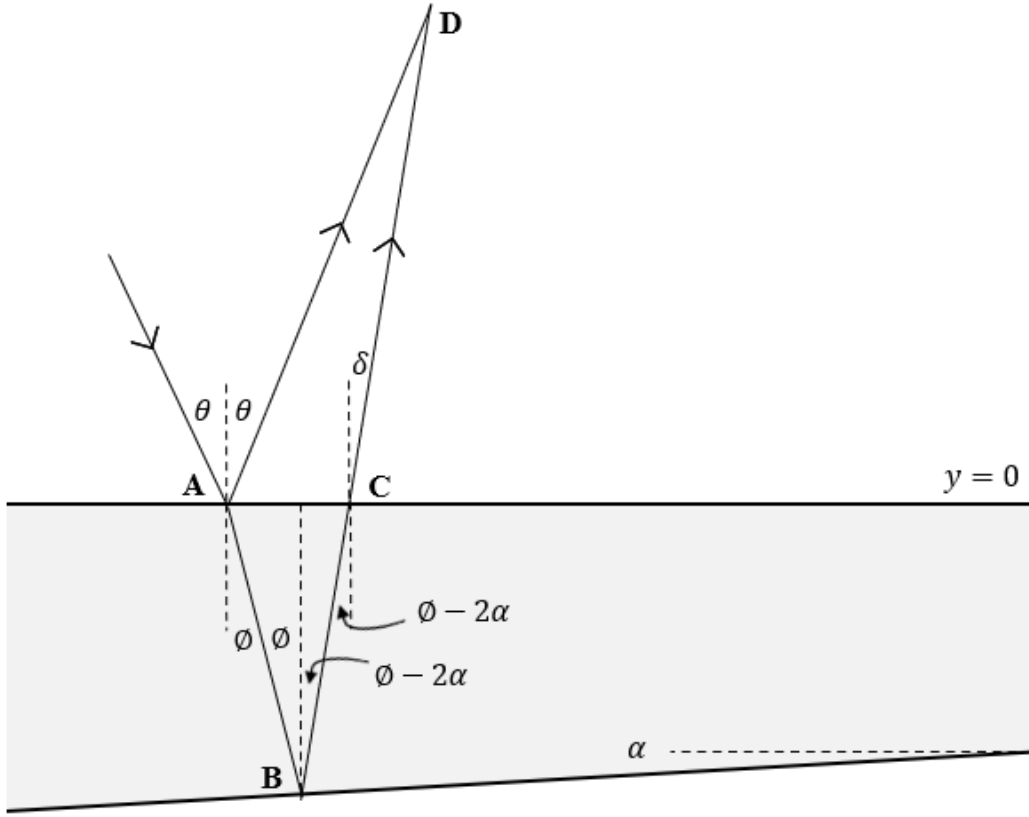


Figure 5. Optical ray model for laser beam hitting Fizeau wedge (grey medium).

The local thickness of the wedge as a function of position x along the wedge is given by:

$$d(x) = \alpha x + t \quad (6)$$

where α is the wedge angle and t is the maximum wedge thickness.

The ray path is divided into four segments d_1 (AB, the forward path of the transmitted ray inside the wedge), d_2 (BC, the return path of the transmitted ray inside the wedge), d_3 (AD, the air path of the reflected ray to the intersection), and d_4 (CD, the air path of the transmitted ray to the intersection). These segments are expressed as:

$$d_1 = \frac{d(x)}{\cos(\phi)} \quad (7)$$

$$d_2 = \frac{d(x)}{\cos(\phi - 2\alpha)} \quad (8)$$

$$d_3 = \sqrt{y_i^2 + (x_i - d(x) \tan(\phi))^2} \quad (9)$$

$$d_4 = \sqrt{y_i^2 + (x_i - d(x) \tan(\phi - 2\alpha))^2} \quad (10)$$

Where (x_i, y_i) are the coordinates of the intersection point of the reflected and transmitted rays relative to point A.

The angles ϕ and δ are related to the incidence angle θ by Snell's law:

$$\phi = \arcsin\left(\frac{1}{n} \sin(\theta)\right) \quad (11)$$

$$\delta = \arcsin(n \sin(\phi - 2\alpha)) \quad (12)$$

From the geometry, the x-coordinates of the reflected and transmitted rays at the wedge surface ($y = 0$) are:

$$x_{r0} = -d(x) \tan(\phi) \quad (13)$$

$$x_{t0} = d(x) \tan(\phi - 2\alpha) \quad (14)$$

The lines describing the reflected and transmitted rays are:

$$y_r = \tan\left(\frac{\pi}{2} - \theta\right) + d(x) \tan\left(\frac{\pi}{2} - \theta\right) \tan(\phi) \quad (15)$$

$$y_t = \tan\left(\frac{\pi}{2} - \delta\right) - d(x) \tan\left(\frac{\pi}{2} - \delta\right) \tan(\phi - 2\alpha) \quad (16)$$

Solving this system of equations gives the coordinates of intersection (x_i, y_i) , and the distance from point A to the intersection is:

$$D = \sqrt{x_i^2 + y_i^2} \quad (17)$$

Finally, the phase difference at the intersection point is calculated using the optical path lengths of the reflected and transmitted rays:

$$\begin{aligned}
 d_r &= d_3 - d_4 \\
 d_t &= (d_1 + d_2) n \\
 \Delta\phi &= \frac{2\pi}{\lambda} (d_r + d_t)
 \end{aligned} \tag{18}$$

Where λ is the wavelength of the laser.

4.2 Python Modelling

4.2.1 Zero-Shear Plane Modelling

Python modelling was used to determine both the distance to the zero-shear plane and the phase difference between the reflected and transmitted rays along a 1" (25.4 mm) Fizeau wedge. The calculation accounts for the laser's incident angle θ , wedge angle α , refractive index n , and local wedge thickness. Rays were modeled as straight lines in air with refraction inside the wedge, and their intersection defines the zero-shear plane. The resulting distance, D , computed along the wedge, is shown in Figure 6, illustrating how the zero-shear plane shifts across the wedge.

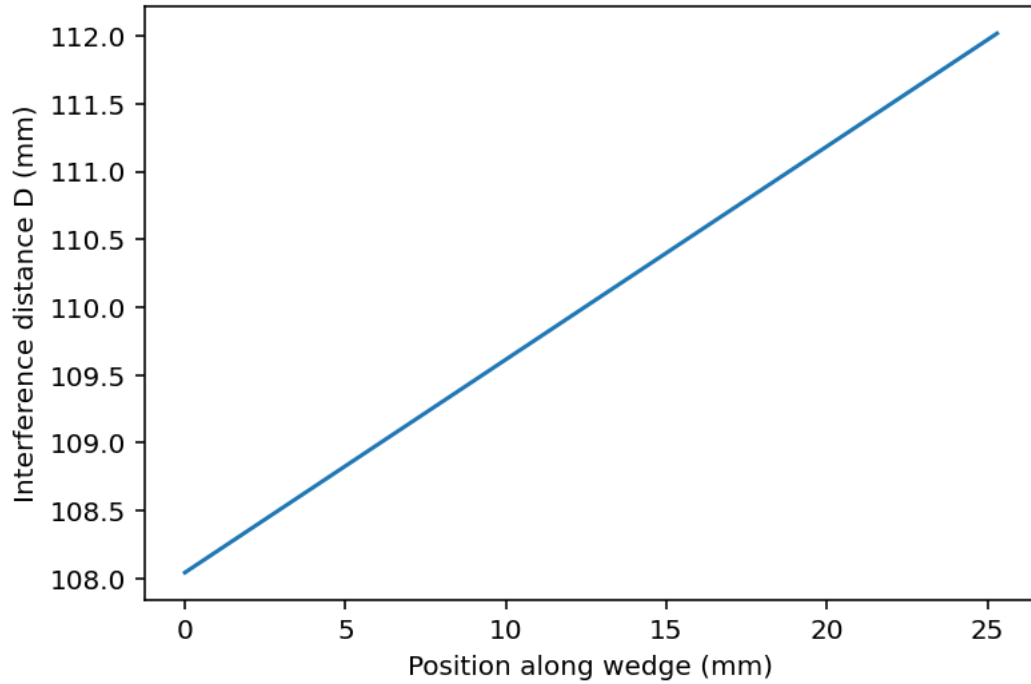


Figure 6. Distance to the zero-shear plane as a function of position along a 1" Fizeau wedge for an incidence angle of 22° .

The phase difference $\Delta\phi$ between the reflected and transmitted rays at their intersection was calculated from the OPD for a laser wavelength of $\lambda = 532 \text{ nm}$. The resulting phase, shown in Figure 7 highlights regions of constructive and destructive interference corresponding to the expected fringe maxima and minima, providing guidance for optimal sensor placement.

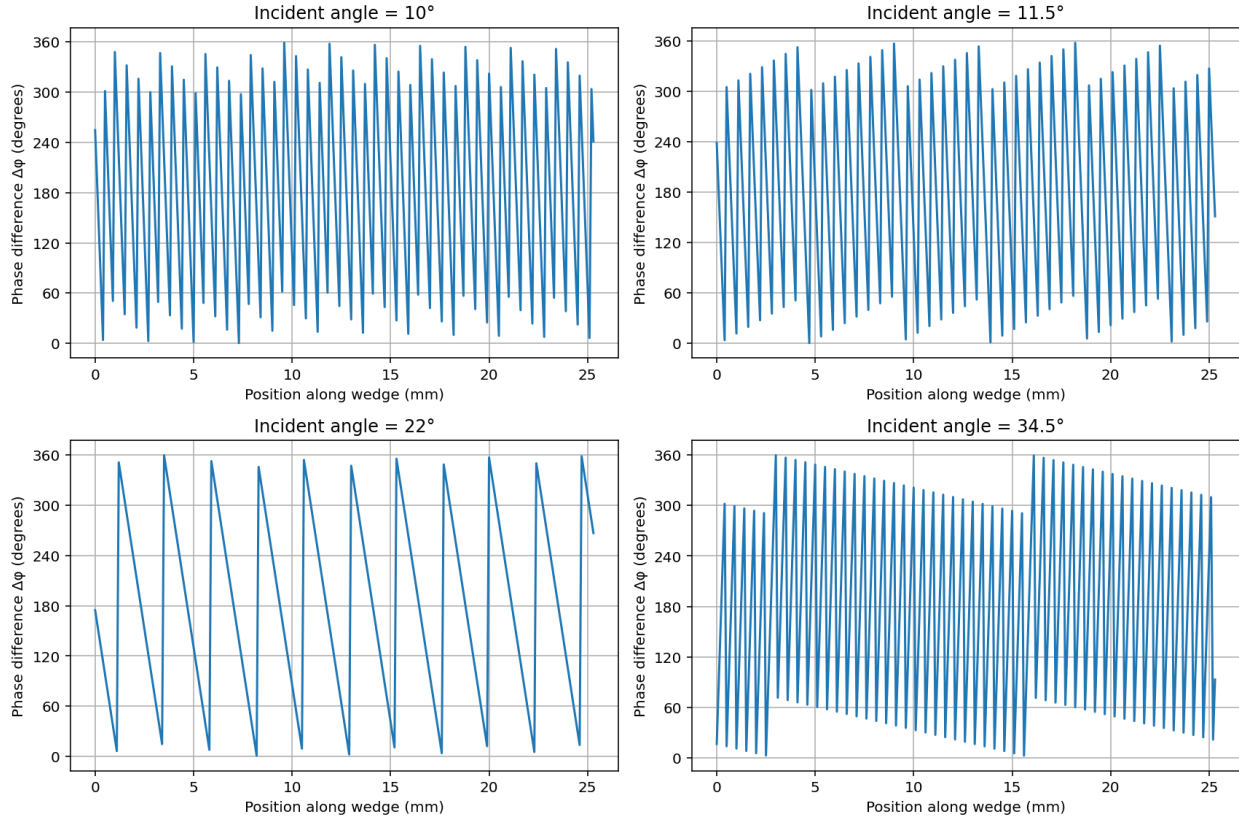


Figure 7. Phase difference at the zero-shear plane along a 1" wedge for a 532 nm laser. Each subplot shows the phase difference for a different incident angle: 10°, 11.5°, 22°, and 34.5°. Significant variations in the phase pattern are observed across different angles, highlighting the critical role of the incident angle in capturing a representative interference pattern.

4.2.2 Wavelength Determination using FFT Simulations

To validate the computational approach used for wavelength extraction, simulations were performed to generate an interference pattern and evaluate the Python analysis pipeline. These simulations were conducted to verify the effectiveness of the fringe analysis method before applying it to experimentally acquired data.

A two-dimensional (2D) interference pattern corresponding to a wavelength of 532 nm and wedge angle of 30 arcmin was numerically generated based on the expected fringe spacing for the Fizeau wedge geometry. The simulated pattern reproduces the spatial variation in intensity associated with the interference of two reflected beams. As shown in Figure 8, a 1D slice is taken

through the center of the fringe pattern, perpendicular to the fringes, to extract an intensity profile for frequency-domain analysis.

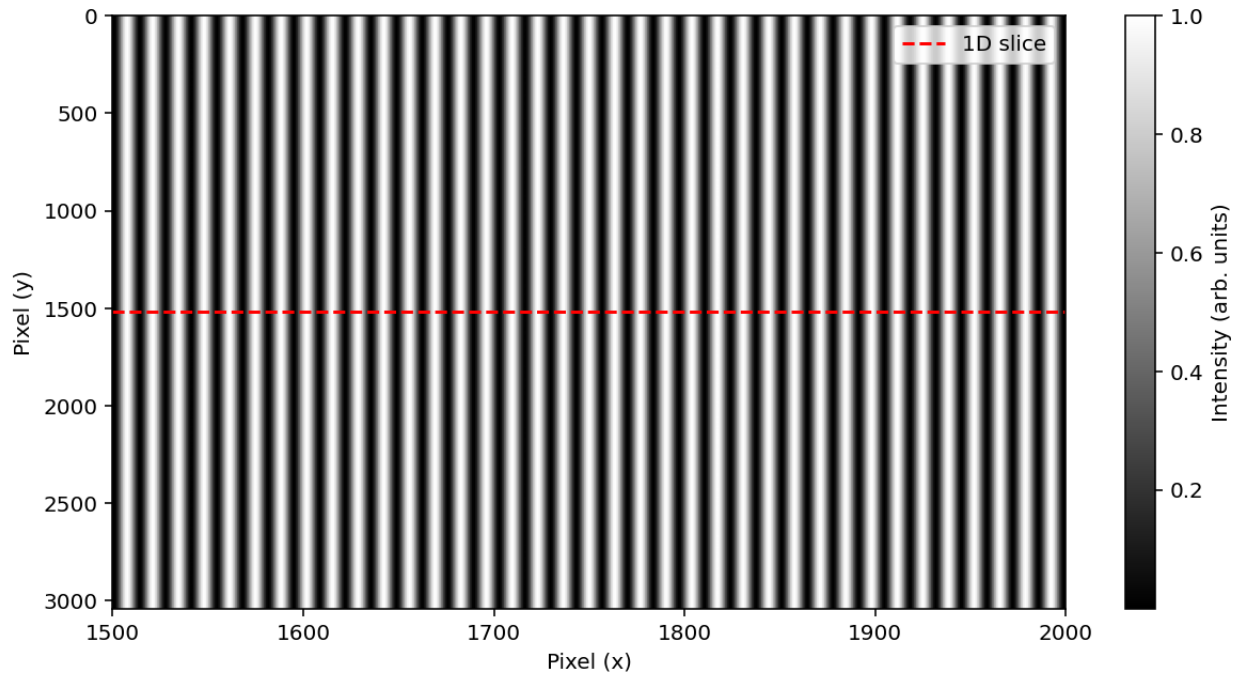


Figure 8. Simulated interference pattern for a wavelength of 532 nm and a wedge angle of 30 arcmin generated using a Fizeau wedge model. The image is shown over a restricted pixel range to highlight the fringe structure. The dashed red line indicates the location of the 1D slice used to extract the fringe intensity profile.

Figure 9 shows the 1D fringe intensity profile with its corresponding FFT of the fringe profile from the 2D interference pattern. The resulting spectrum displays a clear dominant frequency peak, corresponding to the fringe spacing in the spatial domain. The simulated data represents an idealized case and does not capture noise, intensity variations, or optical imperfections present in experimental measurements. As a result, the FFT exhibits a single dominant frequency peak, whereas real experimental data is expected to produce additional frequency components due to intensity variations.

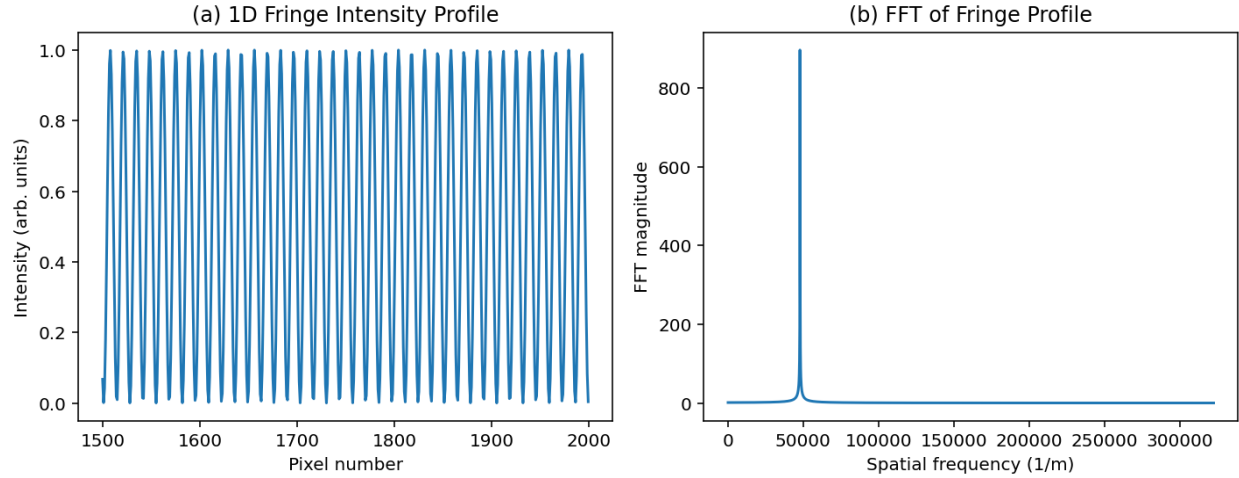


Figure 9. (a) 1D intensity profile extracted from the simulated interference pattern shown in Figure 7. The periodic variation corresponds to the interference fringes. (b) FFT of the intensity profile, showing a dominant spatial frequency peak corresponding to the fringe spacing. The data is shown over a restricted pixel range for clarity.

Identification of this dominant spatial frequency demonstrates the feasibility of using an FFT approach to recover fringe spacing and, by extension, the incident wavelength. Applying this method to the simulated data yields a recovered wavelength of 532 nm, in agreement with the input wavelength used to generate the interference pattern [16]. These results confirmed that the proposed Python Analysis pipeline is capable of accurately resolving the laser wavelength.

5 Final Design

5.1 Overview and Apparatus

The final wavemeter design consists of an integrated optical and computational system combining a Fizeau wedge interferometer, a CMOS camera, and a Raspberry Pi 5. A collimated laser beam is directed onto a 30 arcmin fused silica wedge, where interference fringes are generated due to the OPD between reflections from the upper and lower surfaces. These fringes are captured by a camera positioned at the zero-shear plane and processed in real-time to extract wavelength information to be displayed.

The computational component of the system is implemented on a Raspberry Pi 5, which performs image acquisition, signal processing, and wavelength extraction using an FFT Python analysis pipeline. The Python analysis pipeline can be found in Appendix A and follows the logic flow described by Figure 10.

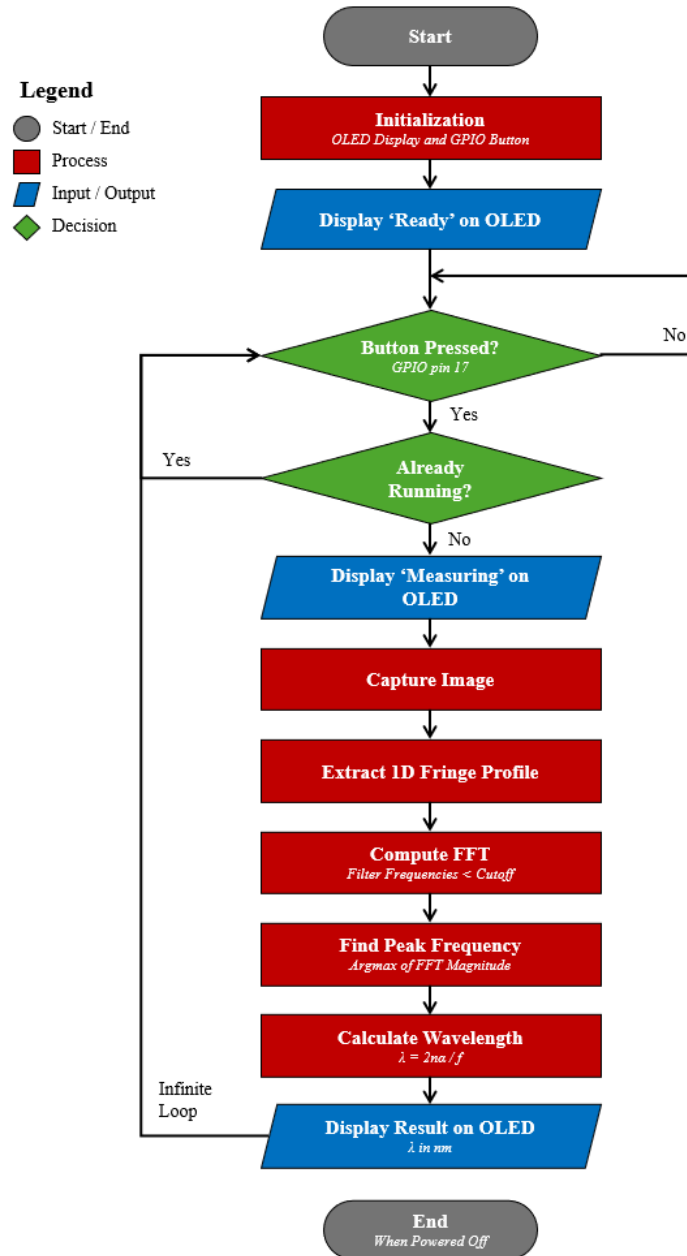


Figure 10. Flowchart of the Fizeau wedge Python analysis pipeline running on the Raspberry Pi. On startup, the system initializes the OLED display and GPIO button, then enters an infinite loop

awaiting user input. Upon a button press, an image is captured via the Pi camera, the central row is extracted as a 1D intensity profile, and an FFT is performed to identify the dominant spatial frequency of the fringe pattern. The recovered wavelength is computed and displayed on the OLED Display in nanometers. The system loops continuously until it is powered off.

The recovered wavelength is displayed on a mini I²C OLED display, enabling real-time feedback during operation. The system is designed to operate as a compact, standalone instrument, controlled via a single push-button to trigger measurements, with SSH access used for initial setup and extracting images and plots. The physical implementation of the wavemeter is assembled on an 8" × 8" optical breadboard and includes the optical, mechanical, and electronic components required for operation. A side view of the complete apparatus is shown in Figure 11.

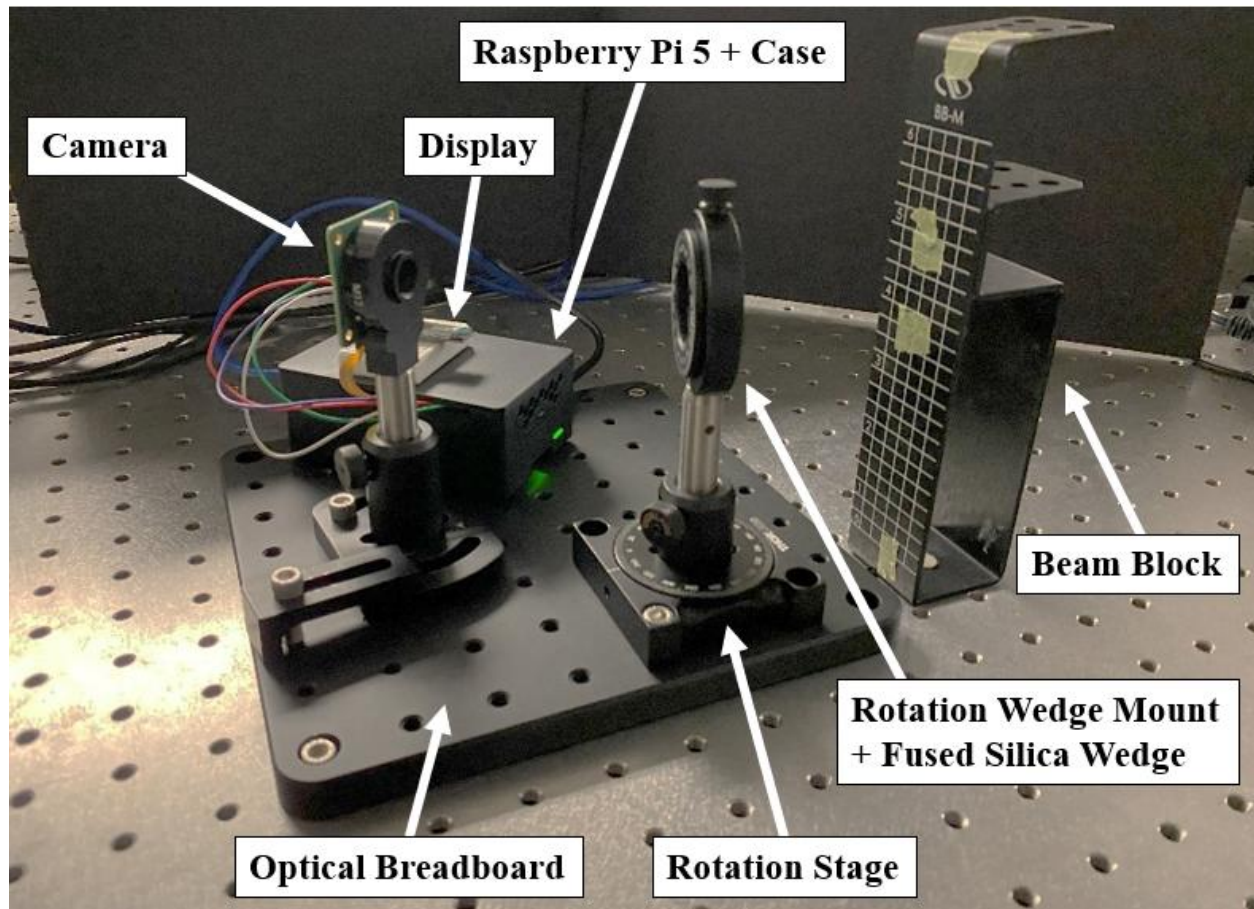


Figure 11. Side view of the complete wavemeter apparatus assembled on an 8" × 8" optical breadboard. Major components are labeled, including the CMOS camera, OLED display (button not pictured, but is beside the display), Raspberry Pi 5 housed in an enclosure, beam block for

optical safety, rotation stage, rotation mount containing the fused silica Fizeau wedge, and the optical breadboard.

The system includes a Raspberry Pi HQ CMOS camera for fringe imaging, a Raspberry Pi 5 housed in an enclosure for data acquisition and processing, a mini PC OLED display for wavelength readout, a fused silica Fizeau wedge mounted on a rotation stage for precise angular alignment and various optomechanical posts and post-holders. A beam block safely terminates the refracted beam, and supporting optical hardware ensures proper alignment between the laser, wedge, and imaging system.

Figure 12 provides a detailed view of the Raspberry Pi enclosure with the lid removed, showing the internal connections required for operation, including the camera ribbon cable, GPIO connections to the OLED display (button GPIO covered in picture), Ethernet connection for SSH communication (required when working over school Wi-Fi because of multifactor authentication), and the power supply.

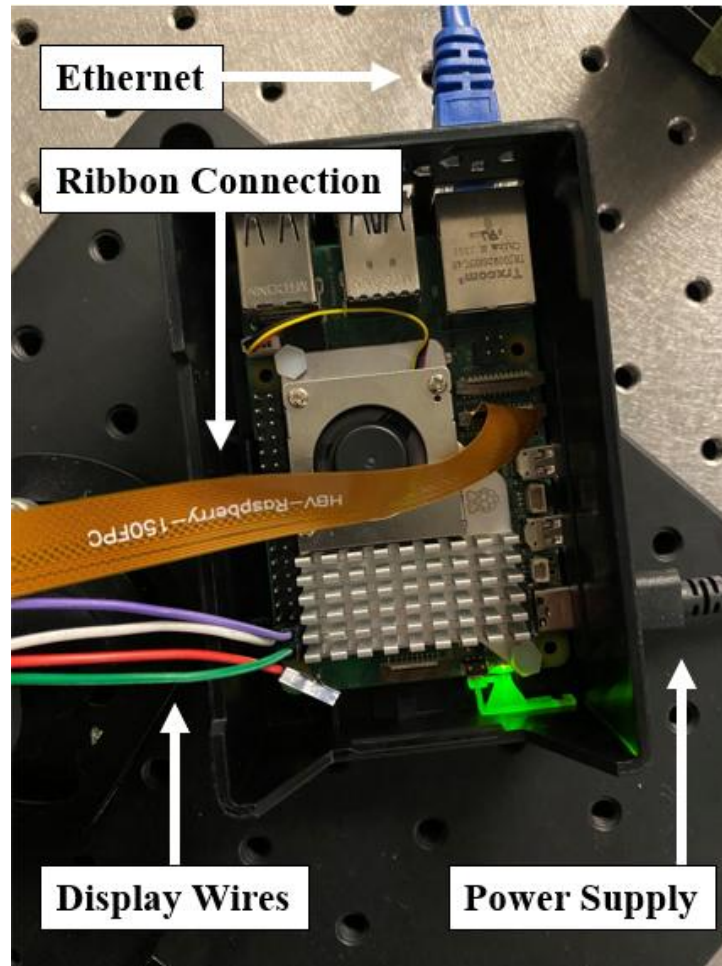


Figure 12. Internal view of the Raspberry Pi 5 enclosure with the lid removed, showing the ribbon cable connection to the camera, GPIO jumper connections to the OLED display, Ethernet cable used for SSH communication and the power supply connection to the Raspberry Pi.

A top-down view of the optical layout is shown in Figure 13, illustrating the laser beam path through the system and its alignment with the Fizeau wedge and imaging sensor.

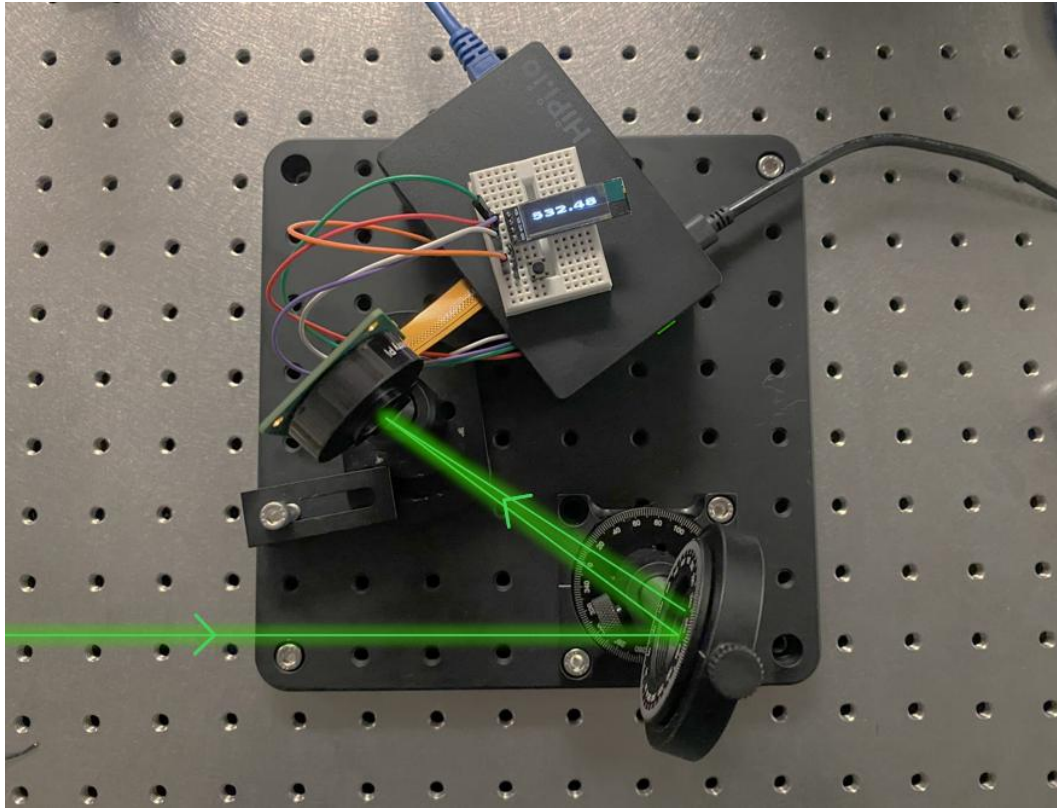


Figure 13. Top view of the optical setup illustrating the laser beam path through the system, including alignment with the Fizeau wedge. Additionally, the mini breadboard with the OLED display and button can be observed.

5.2 Key Decision Making

5.2.1 Camera Selection

The choice of camera was strongly influenced by the simulated interference patterns and the expected fringe spacing across the Fizeau wedge. The Python simulations (Section 4.2.2) demonstrated that the interference fringes produce a spatial frequency that must be adequately sampled to recover wavelengths using an FFT.

To ensure accurate reconstruction of the fringe pattern, the camera required sufficient pixel density across the region of interest such that multiple pixels sample each fringe stripe. This requirement follows directly from the need to resolve the dominant spatial frequency observed in the simulated FFT (Figure 9). A CMOS camera was selected due to its cost-effectiveness and

sensitivity over the wavelength range of interest. The chosen Raspberry Pi HQ camera sensor provides approximately seven pixels per fringe stripe; ensuring that the fringe spacing is captured with sufficient detail for reliable frequency-domain analysis.

Additionally, the selected camera can be operated in a NoIR configuration by removing the infrared filter, enabling extended sensitivity across a broader wavelength range if required.

5.2.2 Incidence Angle and Zero-Shear Plane Position

The incidence angle of the incoming beam is a critical parameter in determining the location of the zero-shear plane and the quality of the interference pattern. The optical ray model developed in Section 4.1 was used to calculate the intersection of the reflected and transmitted rays and predict how the zero-shear plane varies with incidence angle and wedge geometry.

Simulation results (Figure 6) showed that the distance to the zero-shear plane has a small variation depending on the lasers incidence location along the wedge. These results were used to inform the placement of the camera. Furthermore, the phase difference distributions (Figure 7) indicated that different angles produce different fringe characteristics and spatial distributions.

Based on these results, an incidence angle of 22° was selected. This value represents a balance between positioning the zero-shear plane within the spatial constraints of the $8'' \times 8''$ optical breadboard and maintaining a fringe pattern with sufficient spatial variation for analysis.

5.2.3 Computational Approach

The computational approach implemented on the Raspberry Pi 5 was also motivated by the modelling and simulation results. The FFT Python analysis pipeline for wavelength extraction was validated in Section 4.2.2 using synthetic interference data, where a dominant spatial frequency peak corresponding to the known input wavelength was successfully recovered.

This validation justified the use of this analysis pipeline for processing experimental fringe data. The Raspberry Pi 5 was selected as the processing platform due to its ability to perform real-time image acquisition and FFT computations while maintaining a compact and low-cost system.

5.3 Iterations

Several design iterations were undertaken to refine the final system configuration. The initial design included caged irises for beam alignment and leveling. These components were ultimately removed, as beam leveling was deemed outside the scope of this project. The wavemeter is intended for use inline with existing experimental setups, where beam alignment is typically already established. In the final implementation environment, the incoming beam was sufficiently level, and additional alignment components were unnecessary.

Early testing was conducted without a Gaussian beam profile, which resulted in non-uniform intensity distributions across the sensor. This issue was resolved by using a well-collimated Gaussian beam, which produced a more uniform and representative intensity profile.

An initial imaging setup using the Raspberry Pi Module 3 Camera with autofocus was also evaluated. However, the autofocus consistently focused at longer distances, preventing clear imaging of the interference fringes. This limitation made it unsuitable for the application, and it was replaced with a Raspberry Pi HQ Camera M12, which allowed for proper manual focusing and successful fringe capture.

5.4 Economic Analysis

Table 1 summarizes the primary components required for the construction of the wavemeter, along with their intended functions and associated costs. The total cost of the system is \$972.87 CAD, which remains well below the target budget of \$1500 CAD.

Table 2. Bill of materials for the final design Fizeau wedge wavemeter.

Component	Function	Qty.	Unit Cost (CAD, pre-tax)	Source
Raspberry Pi HQ Camera M12	Captures the interference pattern.	1	\$72.45	[17]

Component	Function	Qty.	Unit Cost (CAD, pre-tax)	Source
Raspberry Pi 5 Starter Kit - 4GB	Processes the data captured by the camera. Performs the computational analysis and extracts the wavelength.	1	\$194.95	[18]
Raspberry Pi Camera Cable Standard - Mini - 200mm V2	Connects the standard MIPI ribbon connection on the camera to the smaller connector on the Pi5.	1	\$2.95	[19]
0.91 Inch White OLED Display Module I2C Interface	Displays the output wavelength computed by the Pi5	1	\$5.95	[20]
Optical Breadboard Plate, Solid Aluminum, 8 x 8 in., 1/4-20, Double Density	Provides a rigid platform for mounting all optical components.	1	\$152	[21]
30 Arcmin Fused Silica 1" Wedge	Fizeau wedge required to produce interference fringes.	1	\$118	[22]
RSP1 Rotation Mount for Ø1" (25.4 mm) Optics	Holds and allows precise angular adjustment of for aligning the Fizeau wedge.	1	\$105.31	[23]

Component	Function	Qty.	Unit Cost (CAD, pre-tax)	Source
RP01 Ø2" Manual Rotation Stage	Provides fine rotational control for precise alignment relative to the incident beam.	1	\$119.18	[24]
TR2 Ø1/2" Optical Post, SS, 8-32 Setscrew, 1/4"-20 Tap	Provides mounting support for optical components at adjustable heights.	2	\$6.31	[25]
UPH2 Ø1/2" Universal Post Holder, Spring Loaded Locking Thumbscrew	Secures optical posts to the breadboard and enables stable positioning.	2	\$38.77	[26]
Subtotal:			\$860.95	
Taxes (13%):			\$111.92	
Total:			\$972.87	

Components associated with beam preparation, including the laser source and optical elements required to produce a well-collimated Gaussian beam, are not included in the financial analysis. The present design assumes an ideal, collimated Gaussian input beam and therefore does not account for the cost of beam generation or conditioning equipment.

6 Results

The wavemeter was tested using two laser sources: a frequency-doubled Neodymium-doped Yttrium Aluminum Garnet (Nd:YAG) green laser with a theoretical wavelength of 532 nm [16], and a red diode laser independently characterized by spectrometer prior to measurement. For

each source, the full-field fringe image, one-dimensional intensity profile, and corresponding FFT are presented. The effect of laser power on camera saturation is also investigated.

6.1 Green Nd:YAG Laser (532 nm)

Figure 14 shows a close-up of the fringe pattern, in which the alternating bright and dark bands are clearly resolved, confirming sufficient beam pixel resolution for accurate wavelength extraction.

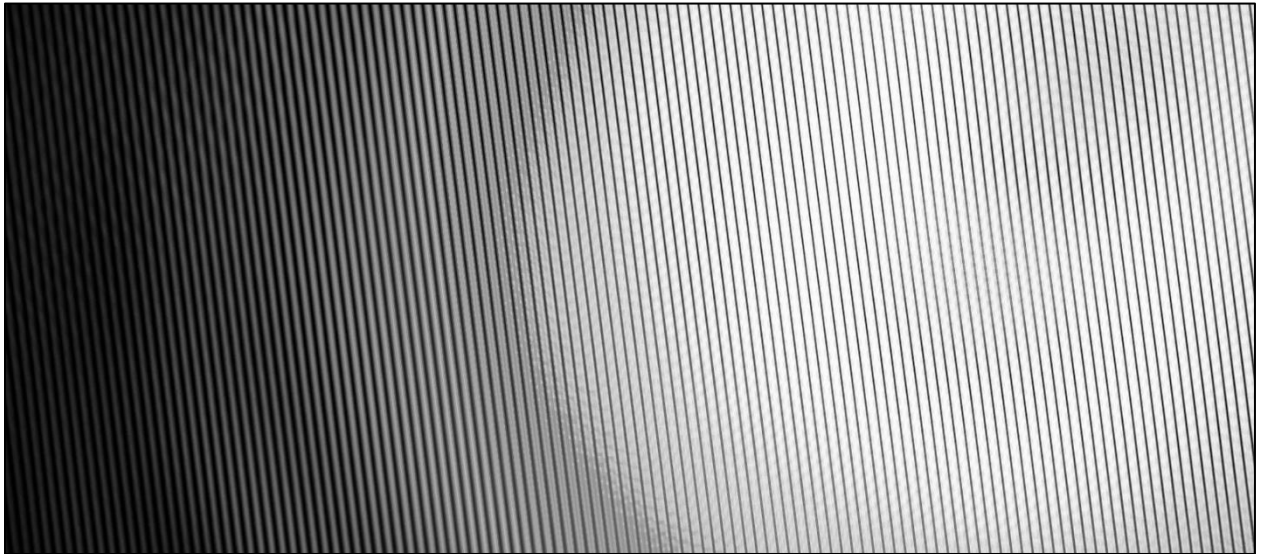


Figure 14. Close-up of the interference fringe pattern produced by the green laser, showing clearly resolved, high-contrast fringes.

Figure 15 shows the full-field fringe image across the sensor, from which a one-dimensional intensity slice was extracted for analysis.



Figure 15. Full-field interference fringe image for the green laser. Captured at 1 mW, an exposure of 500 ms and index of refraction of 1.4607.

The extracted profile and its FFT are shown in Figure 16. The FFT yields a dominant spatial frequency peak corresponding to the fringe spacing, from which a measured wavelength of 532.48 nm is obtained.

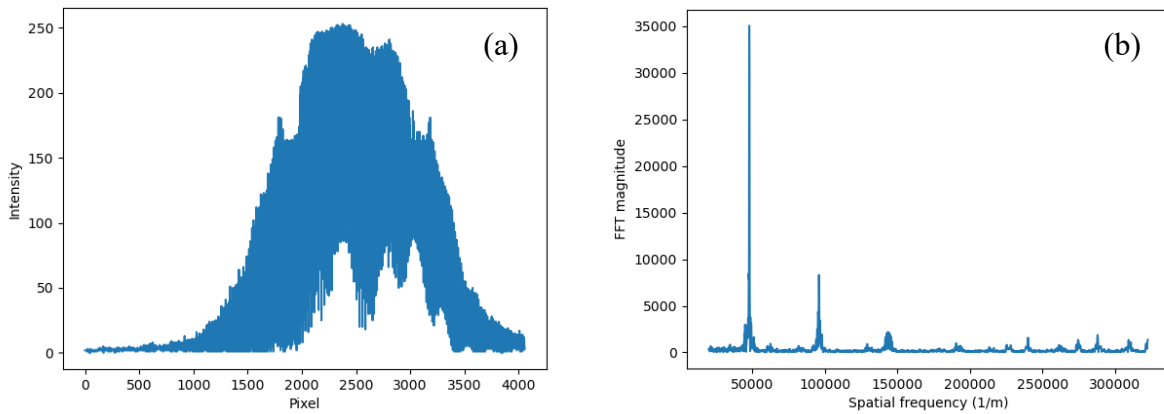


Figure 16. (a) 1D intensity profile extracted from the green laser fringe image (8-bit grayscale pixel intensity, 0–255). (b) FFT of the profile, with the dominant peak used to determine the fringe spacing and wavelength.

6.2 Red Diode Laser (653 nm)

The wavelength was first measured using a spectrometer (Figure 17). A single peak is observed at 653.068 ± 0.001 nm, which is the reference value for comparison.

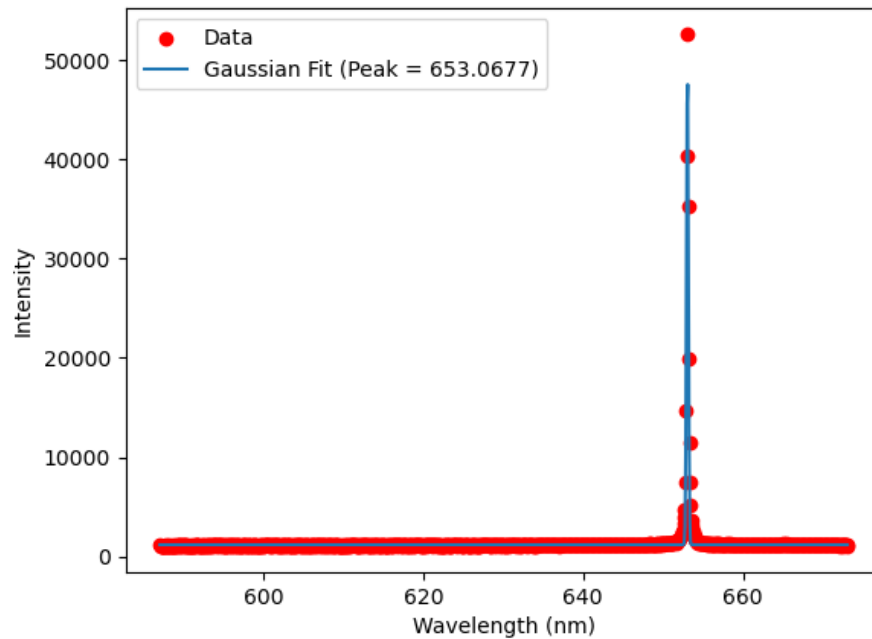


Figure 17. Spectrometer measurement of the red diode laser, showing a single emission peak at 653 nm. This value is used as the reference wavelength for comparison with the Fizeau wavemeter result.

The laser was then measured with the wavemeter. Figure 18 shows the full-field fringe image. A higher exposure rate was used to capture the fringe image.

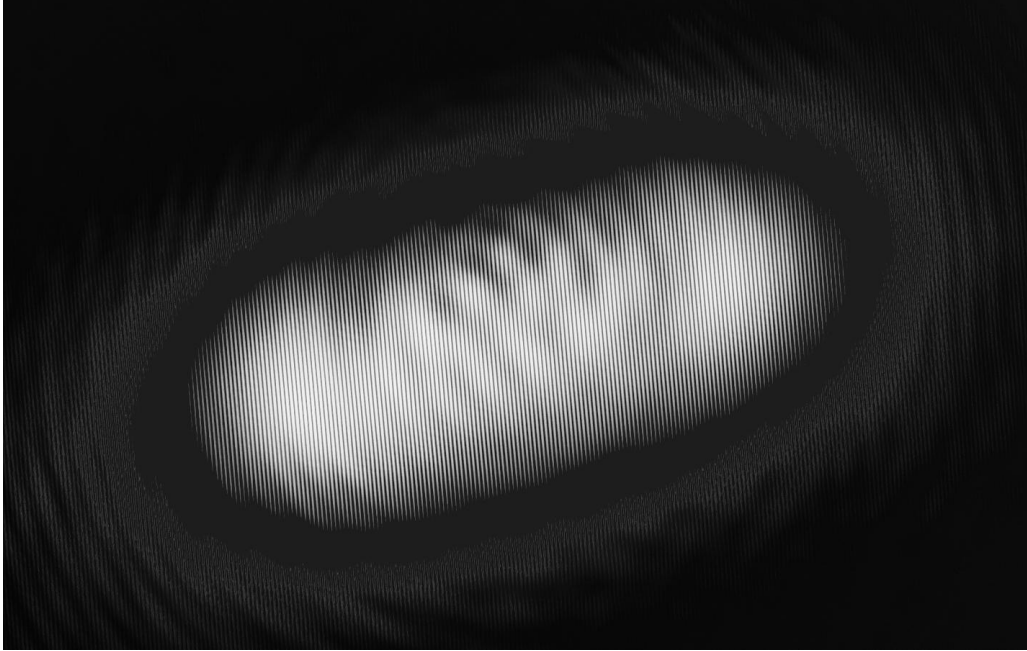


Figure 18. Full-field fringe image for the red diode laser. A wider interference pattern is the result of not using an iris and spatial filter to prepare the beam before measurement. Captured at 44.28 mW, an exposure of 3000 ms and index of refraction of 1.4565.

The 1D profile and FFT are shown in Figure 19. The dominant FFT peak occurs at a lower spatial frequency than for the green laser, reflecting the longer fringe spacing. The extracted fringe spacing yields a measured wavelength of 654.19 nm.

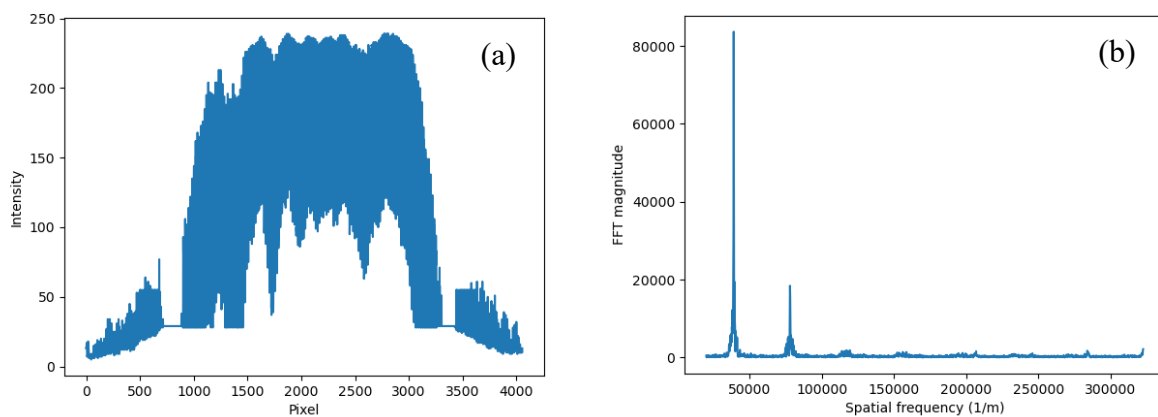


Figure 19. (a) 1D intensity profile extracted from the red laser fringe image (8-bit grayscale pixel intensity, 0–255). (b) FFT of the profile, with the dominant peak used to determine the fringe spacing and wavelength.

6.3 Sensor Saturation

When incident power is too high, pixel saturation causes the intensity profile to clip, distorting the fringe shape and introducing error in the FFT-derived wavelength. This was investigated using the green laser at three power levels, with the resulting 1D profiles shown side by side in Figure 20.

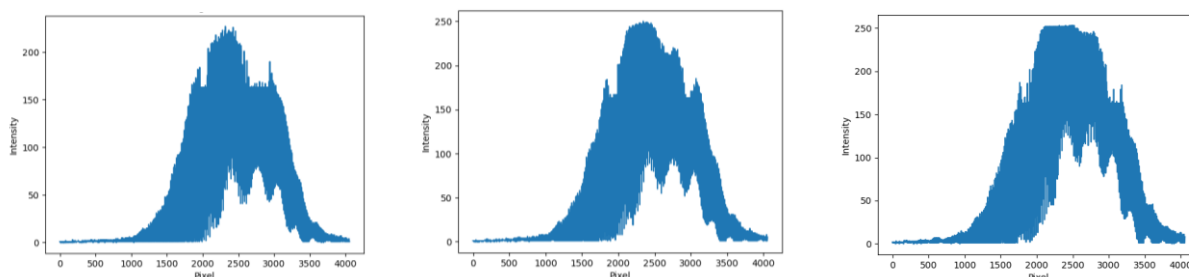


Figure 20. 1D intensity profiles for the 532 nm laser at three power levels (8-bit grayscale pixel intensity, 0–255). (Left) 0.5 mW: well-defined sinusoidal fringes. (Centre) 1 mW: peaks approaching saturation. (Right) 1.5 mW: clipped profile, indicating sensor saturation. Higher-power lasers require attenuation for use with this apparatus.

7 Discussion

The wavemeter successfully recovered the wavelength of both laser sources. For the green Nd:YAG laser, the measured wavelength of 532.48 nm compares to the theoretical value of 532 nm, a discrepancy of 0.48 nm. For the red diode laser, the measured wavelength of 654.19 nm compares to the spectrometer reference of 653.07 nm, a discrepancy of 1.12 nm. Without calibration, the green laser result approaches the ± 0.1 nm design target, while the red laser exceeds it

These results also confirm that, for the green laser, powers exceedingly approximately 1 mW saturate the camera sensor for this apparatus, and optical attenuation, such as a neutral density filter, is required for higher-power sources at this wavelength.

7.1 Sources of Uncertainty

7.1.1 Pre-Calibration Systematic Uncertainties

The manufacturer specifies a nominal pixel pitch of 1.55 μm [17]; however, individual sensors may exhibit deviations due to fabrication tolerances. Since fringe spacing is measured in pixels, any error in the assumed pixel size propagates multiplicatively into the recovered wavelength.

For the green laser, the comparison between the measured wavelength of 532.48 nm and the known 532 nm corresponds to a discrepancy of 0.09%, implying an effective pixel pitch of approximately 1.5486 μm . For the red laser, the measured wavelength of 654.19 nm and the known 653.07 nm corresponds to a discrepancy of 0.17%, implying an effective pixel pitch of approximately 1.5473 μm .

This is consistent with rounding of the pixel pitch specification and falls within typical manufacturing tolerances. This uncertainty is removed through calibration against a known reference wavelength, which corrects the effective pixel size.

A second source of uncertainty arises from the wedge angle. The fused silica wedge has a nominal angle of 30 arcmin with a manufacturer-specified tolerance of ± 15 arcmin [22]. Since the recovered wavelength depends linearly on the wedge angle via:

$$\lambda = \frac{2n\alpha}{\Delta f} \quad (19)$$

uncertainty in the wedge angle directly translates into proportional uncertainty in wavelength. A 1 arcmin deviation corresponds to approximately 3.3% of the nominal angle, producing an identical error in wavelength ($\approx \pm 17$ nm at 532 nm) prior to calibration. With calibration this error can be absorbed by a calibration constant.

7.1.2 Post-Calibration Random Uncertainties

The FFT resolves spatial frequencies at intervals given by:

$$\Delta f = \frac{f_s}{N} \quad (20)$$

where f_s is the sampling frequency and N is the number of pixels in the extracted 1D intensity slice. In general, the true fringe frequency does not align with a discrete FFT bin, and the frequency estimate is therefore taken as the bin of maximum magnitude. This introduces a quantization error associated with discretization of the frequency axis [27]. The maximum error due to this bin-centering effect is bounded by:

$$\varepsilon = \frac{\Delta f}{2} \quad (21)$$

which occurs when the actual frequency lies midway between two adjacent bins. Without interpolation, this represents a fundamental lower bound on frequency resolution and, consequently, wavelength precision.

Sub-bin interpolation can improve the effective resolution. Parabolic interpolation fits a quadratic function to the three bins surrounding the FFT peak to estimate its location more precisely. With appropriate windowing (e.g., Blackman–Harris–Nuttall), this reduces the maximum error to approximately 3.34% of Δf , corresponding to an improvement of roughly $15\times$ [27]. Gaussian interpolation, which applies parabolic interpolation to the logarithm of the spectrum, further reduces the maximum error to approximately 0.314% of Δf , yielding an improvement of roughly $159\times$ [27].

Another post-calibration source of uncertainty is the lasers incidence angle, which influences the effective fringe geometry recorded on the sensor. Deviations in incidence angle alter fringe spacing and can shift the zero-shear plane relative to the detector, reducing fringe contrast and introducing measurement error.

This uncertainty is session dependent. When the optical components remain fixed, alignment errors remain constant and are effectively absorbed during calibration. However, if the setup is removed or disturbed, the incidence angle may change, reversing prior calibration.

7.1.3 Environmental Uncertainties

The model used assumes propagation through vacuum ($n = 1$), whereas in practice the beam travels through air with a refractive index of approximately 1.0003 at standard conditions [28]. This value varies with temperature, pressure, and humidity, introducing a small offset in the computed OPD. For higher-precision applications, this effect should be corrected using the Edlén equation, which relates refractive index to atmospheric conditions [28].

7.2 Limitations and Next Steps

A primary limitation is the design's reliance on calibration against a known reference wavelength. While calibration effectively compensates for systematic uncertainties such as pixel pitch and wedge angle, it reduces the system's functionality. Any change in alignment, particularly the laser incidence angle, can invalidate the calibration and require re-calibration before accurate measurements can be obtained. Another limitation arises from the use of an FFT-based fringe analysis method, which is constrained by discrete sampling.

A key next step is implementing sub-bin frequency estimation within the Python analysis pipeline. Incorporating Gaussian interpolation would improve frequency resolution, enabling more precise wavelength recovery while windowing is likely not necessary as the intensity signals approach zero on both ends of the response.

Additionally, interfacing the Raspberry Pi with Dr. Stotz's LabVIEW experimental environment would enhance the functionality and usability of the wavemeter within his laboratory. This integration could be achieved through an Ethernet connection or by utilizing a LabVIEW SSH library to establish communication with the Raspberry Pi.

8 Conclusions

This project successfully designed and built a low-cost Fizeau wedge wavemeter capable of laser wavelength measurement. The final system was assembled for \$972.87 CAD, well within the \$1500 CAD budget target, demonstrating that precision wavelength measurement is achievable without expensive instrumentation.

Experimental testing with two laser sources yielded measured wavelengths of 532.48 nm (vs. 532 nm theoretical) and 654.19 nm (vs. 653.07 nm spectrometer reference), with discrepancies of 0.48 nm and 1.12 nm respectively. These errors are primarily attributable to pixel pitch uncertainty and wedge angle tolerance, both of which can be eliminated through calibration against a known reference wavelength, providing a clear path to meeting the ± 0.1 nm design target.

The most impactful next step is the implementation of sub-bin FFT interpolation, which would reduce wavelength quantization error. Integration with Dr. Stotz's LabVIEW environment would further extend the instrument's utility as a research tool.

Word Count: 4998

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Appendix A. Python Analysis Pipeline

```
# =====  
  
# Fizeau Wedge Python Analysis Pipeline  
  
# =====  
  
# File: wavemeter.py  
  
# Author: Ross Jones  
  
# Course: ENPH 455  
  
# Directory: /home/hellorossj/Wavemeter/wavemeter.py  
  
# Username for SSH: hellorossj  
  
# Password for SSH: Rossthesis  
  
# =====  
  
  
from picamera2 import Picamera2  
  
import cv2  
  
import numpy as np  
  
import matplotlib.pyplot as plt  
  
import time  
  
from luma.core.interface.serial import i2c  
  
from luma.oled.device import ssd1306  
  
from luma.core.render import canvas  
  
from PIL import ImageFont  
  
from gpiozero import Button  
  
from signal import pause  
  
running = False  
  
  
# =====  
  
# 1. PARAMETERS  
  
# =====
```



```

n = 1.4607          # refractive index (1.4607 for 532 nm, 1.4565 for 653 nm)

alphaDeg = 0.5      # wedge angle (deg) - corresponds to 30 arcmin

alpha = np.radians(alphaDeg)

pixel_size = 1.55e-6    # pixel size (m)

# =====

# 2. CAPTURE IMAGE FROM PI CAMERA

# =====

# Function to capture an image from the Pi Camera and save it as a grayscale PNG
def capture_image(filename="fringe_image.png"):

    picam2 = Picamera2()

    config = picam2.create_still_configuration(main={"format": "RGB888"})

    picam2.configure(config)

    picam2.start()

    picam2.set_controls({

        "AeEnable": False,

        "ExposureTime": 1000, # Adjust as needed for different lasers

        "AnalogueGain": 2.0

    })

    time.sleep(1)

    image = picam2.capture_array()

    picam2.stop()

```

```

picam2.close()

gray_image = cv2.cvtColor(image, cv2.COLOR_RGB2GRAY)

cv2.imwrite(filename, gray_image)

return gray_image

# =====

# 3. PRINTING TO THE DISPLAY

# =====

# Function to display text on the OLED screen
def show_oled_text(text, font_size=30):
    serial = i2c(port=1, address=0x3C)
    device = ssd1306(serial)

    font = ImageFont.truetype(
        "/usr/share/fonts/truetype/dejavu/DejaVuSans-Bold.ttf",
        font_size
    )

    # Center the text on the display
    w, h = font.getbbox(text)[2:]
    x, y = (device.width - w)//2, (device.height - h)//2

    with canvas(device) as draw:
        draw.text((x, y), text, font=font, fill=255)

# =====

```

```

# MAIN WAVEMETER FUNCTION

# =====

# Function to perform the wavemeter measurement and analysis

def wavemeter():

    global running

    if running:

        return

    running = True

    show_oled_text("Measuring", font_size=20)

    print("Running measurement...")

    try:

        # Capture the fringe pattern

        I_2D = capture_image()

        Ny, Nx = I_2D.shape

        # Extract the 1D fringe profile from the center row

        row = Ny // 2

        signal = I_2D[row, :]

        # Plot the 1D fringe profile

        plt.figure()

        plt.plot(signal)

        plt.xlabel("Pixel")

        plt.ylabel("Intensity")

        plt.title("1D Fringe Profile")

```

```

plt.savefig("1Dfringe_profile.png")

plt.close()

# Perform FFT

fft_vals = np.fft.fft(signal)

freqs = np.fft.fftfreq(len(signal), d=pixel_size)

fft_mag = np.abs(fft_vals)

# Filter out low frequencies

min_freq = 2e4

valid = freqs > min_freq

freqs = freqs[valid]

fft_mag = fft_mag[valid]

# Plot the FFT magnitude

plt.figure()

plt.plot(freqs, fft_mag)

plt.xlabel("Spatial frequency (1/m)")

plt.ylabel("FFT magnitude")

plt.title("FFT of Fringe Pattern")

plt.savefig("fft_plot.png")

plt.close()

# Identify the dominant frequency

peak_idx = np.argmax(fft_mag)

dominant_freq = freqs[peak_idx]

# Calculate the wavelength

recovered_wavelength = (2 * n * alpha) / dominant_freq

```

```

wavelength_nm = recovered_wavelength * 1e9

print(f'Recovered wavelength: {wavelength_nm:.2f} nm')

show_oled_text(f'{wavelength_nm:.2f}', font_size=30)


# Wait a few seconds before allowing the next measurement to display result
time.sleep(5)


finally:

    running = False

    print("Done. Waiting for next press...")


# =====

# BUTTON SETUP

# =====


button = Button(17, pull_up=True)

button.when_pressed = wavemeter

print("System ready. Press button to take measurement.")

show_oled_text("Ready", font_size=30)

pause()


# =====

# NOTES:

# If connected through SSH this command kills the current process:

# sudo systemctl stop wavemeter.service

# =====

```